

SNAP-8

Acetyl Octapeptide-3 • Topical Anti-Wrinkle Peptide • Research Overview • July 2026

WHAT IT IS

SNAP-8 (INCI: Acetyl Octapeptide-3) is a synthetic 8-amino-acid peptide developed by Lipotec (now Lubrizol) as an extension of Argireline (Acetyl Hexapeptide-3/8). Sequence: Ac-Glu-Glu-Met-Gln-Arg-Arg-Ala-Asp-NH₂. Molecular weight ~1,075 Da. It is a cosmetic ingredient (not a drug) designed as a topical “Botox-like” alternative that softens expression lines by interfering with the SNARE complex involved in neurotransmitter release at the neuromuscular junction. It is used in serums and creams at typical concentrations of 3–10%.

KEY CHARACTERISTICS

FEATURE	DETAILS
Structure	Octapeptide (8 AA); extended version of Argireline (6 AA)
Mechanism	Competitive inhibition of SNAP-25 in SNARE complex → reduced ACh release → softer muscle contraction
Route	Topical only (serums/creams); not intended for injection
Typical Concentration	3–10% in finished formulations (10% used in key manufacturer studies)
Onset	Gradual; measurable effects often reported after 7–28 days of consistent use
Regulatory Status	Cosmetic ingredient (INCI listed); not FDA-approved as a drug

KEY EVIDENCE

Primary data come from manufacturer-sponsored studies (Lipotec/Lubrizol). In a small 28-day trial (17 female volunteers), a 10% SNAP-8 solution applied twice daily produced a mean wrinkle depth reduction of ~35% around the eyes, with a maximum individual reduction of ~63% (silicon replica / profilometry). Manufacturer comparisons claim SNAP-8 is roughly 30% more potent than Argireline at equivalent concentrations. Independent peer-reviewed human RCTs are essentially absent; most supporting data are industry-generated in vitro or small cosmetic panels. Skin penetration to the neuromuscular junction remains a theoretical limitation for all topical SNARE-inhibiting peptides.

REPORTED EFFECTS & POTENTIAL BENEFITS

- Softens expression lines (crow’s feet, forehead, glabellar) by attenuating (not eliminating) facial muscle contraction.
- Marketed as a milder, reversible, topical alternative to botulinum toxin — preserves natural expression better than injectables.
- Often combined with other peptides (e.g., Argireline, Matrixyl, GHK-Cu) in multi-peptide anti-aging formulas such as Rejuven-8-type creams.
- Effects are gradual and require consistent twice-daily use; results reverse when discontinued.

TYPICAL USE IN COSMETIC FORMULATIONS

Concentration: Most commercial products use 3–10% of a SNAP-8 solution (the pure peptide is present at much lower final levels after dilution).

Application: Clean skin, 1–2 times daily (morning and/or evening). Common on face, especially periorbital and forehead areas. Can be layered under moisturizer or sunscreen.

Stacking: Frequently paired with Argireline (same mechanism, complementary), GHK-Cu (repair/firmness), Matrixyl (collagen signaling), and hydrators. Your Rejuven-8 cream already includes SNAP-8 alongside other actives.

SAFETY PROFILE & CONSIDERATIONS

SNAP-8 is generally well tolerated as a topical cosmetic ingredient. Reported effects are mild and uncommon: transient redness, mild irritation, or dryness in sensitive individuals. No serious systemic adverse events are associated with topical use at cosmetic concentrations. Because it is a large peptide, significant systemic absorption is not expected. Not intended for injection. Avoid broken skin or active dermatitis. As with any new topical, patch-test if you have sensitive skin. Evidence quality is limited (mostly manufacturer data), so expectations should be modest compared with injectables or stronger prescription topicals (retinoids).

RELATION TO ARGIRELINE & OTHER ACTIVES

SNAP-8 is the longer (8-AA) version of Argireline (6-AA). Both target the same SNAP-25 / SNARE pathway; SNAP-8 is claimed to have higher binding affinity and slightly greater in-vitro potency. In practice the difference is modest and both are limited by topical penetration. Botulinum toxin works on the same pathway but by irreversible enzymatic cleavage of SNAP-25 (much stronger, injectable, medical). SNAP-8 is frequently stacked with GHK-Cu (which you already use) for complementary repair + expression-line softening.

DISCLAIMER: For educational and research reference only. SNAP-8 is a cosmetic ingredient, not an FDA-approved drug. Efficacy data are primarily manufacturer-sponsored and limited. Individual results vary. This is not medical or dermatological advice. Consult a qualified professional for personalized skincare recommendations. Information synthesized from public cosmetic research summaries and manufacturer data as of July 2026.